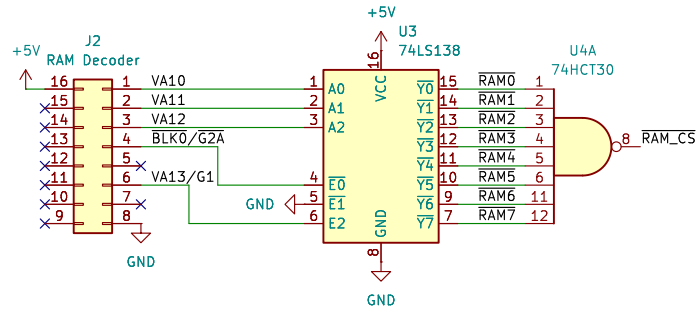
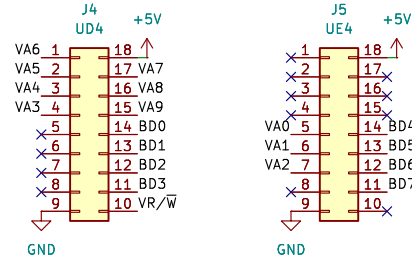


Signals to/from RAM decoder (74LS138)
Original decoder activates internal RAM only for RAM0 and RAM4 to RAM5
We want to activate our replacement for every 1KB chunk in BLK0, including RAM1 to RAM3
This will give us the full 8KB in BLK0 – standard 5KB plus 3KB RAM expansion

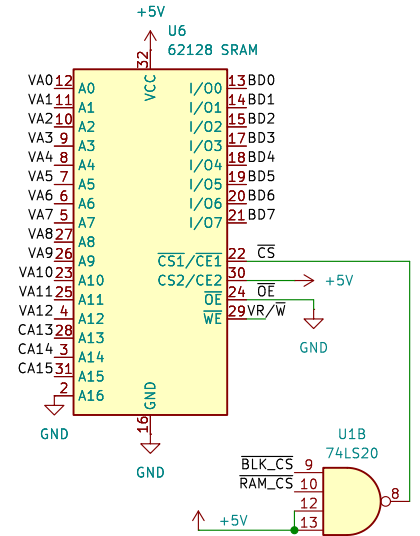


Original 2114 RAM chips are removed from the VIC.
Two are replaced with IC sockets so daughterboard can be installed.

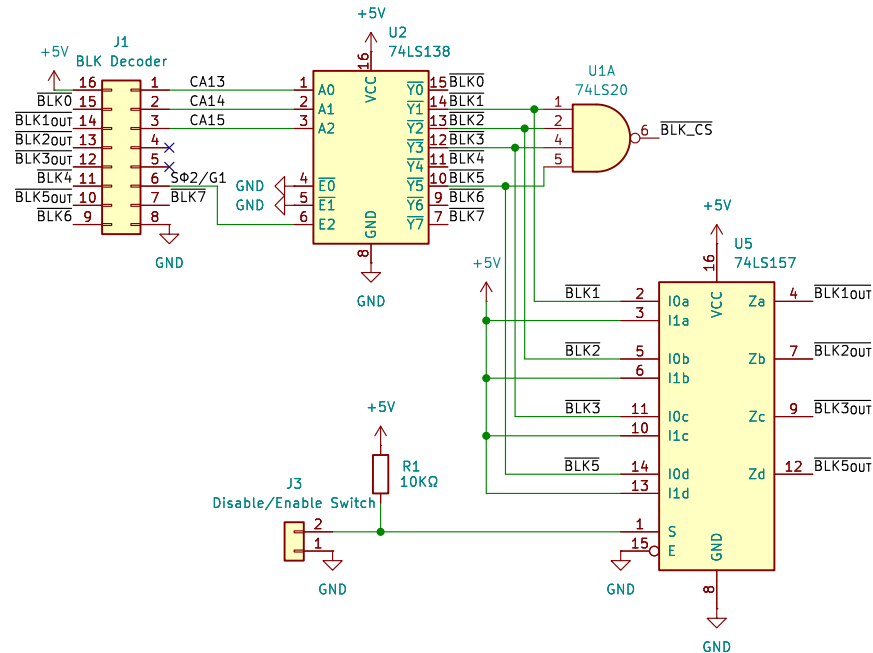


Refer to the "DIP-16 Breakout Board" for the J1 & J2 connection

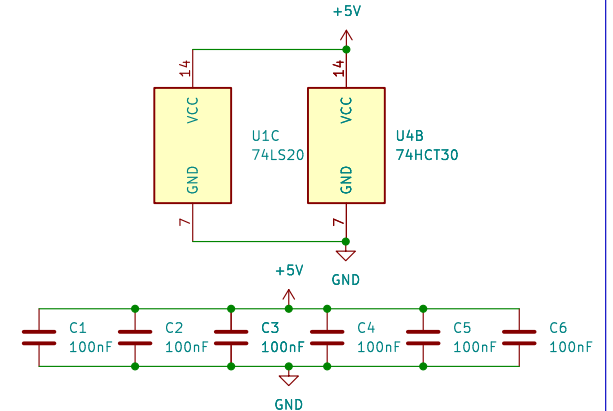
40KB RAM Expansion



Signals to/from BLK decoder (74LS138)
We want to pass BLK0 back so it can be used for BLK0 handling like normal (input to RAM decoder)
We want to capture BLK1 to BLK3 & BLK5 selects and use for our RAM expansion
But we also want to be able to disable the expansion for software compatibility
This can be done via a switch:
* When the switch is high, the BLK1/BLK2/BLK3/BLK5 are passed back to the VIC, disconnecting our RAM expansion from the data bus and passing through to the cartridge port
* When the switch is low, these are blocked which prevents the VIC from disconnecting our RAM from the data bus



Power & Decoupling



www.youtube.com/@Brfff

Brett Hallen

Sheet: /

File: VIC-20_Internal_40KB_RAM.kicad_sch

Title: Commodore VIC-20 40KB RAM Expansion

Size: A4 Date: 27/MAY/2026

KiCad E.D.A. 9.0.9

Rev: A

Id: 1/1